Chinese Remainder Theorem

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https://soham-saha.github.io

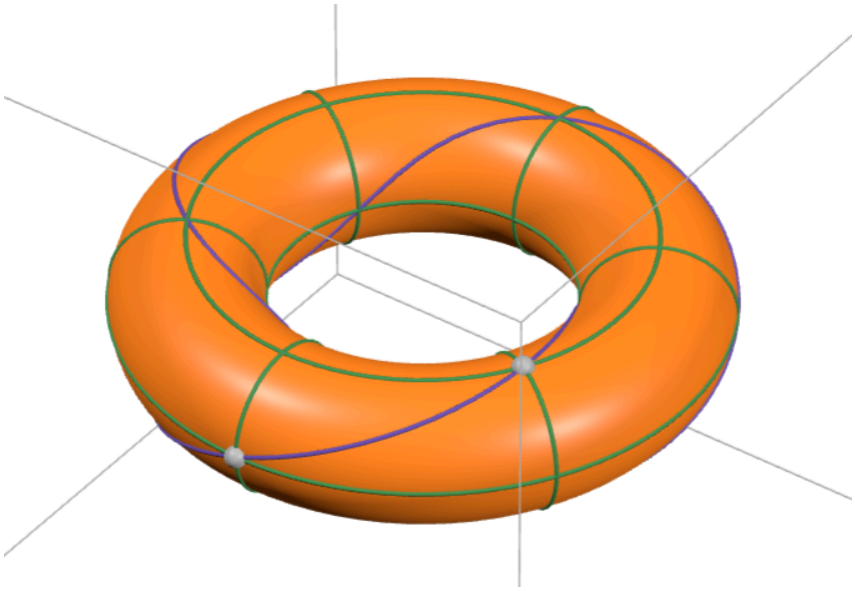
https://soham-saha.codeberg.page

Given integers m, n, the Chinese Remainder Theorem states that the following are equivalent:

- $\gcd(m, n) = 1$ (i.e., m and n are coprime).
- Given integers a, b, there exists exactly one integer x such that $0 \leq x < mn$, $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$.
- $\mathbb{Z}/(mn)\mathbb{Z}$ and $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ are isomorphic as groups. In fact, the homomorphism $f : \mathbb{Z}/(mn)\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ induced by the homomorphisms $\pi_m^{mn} : \mathbb{Z}/(mn)\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$ and $\pi_n^{mn} : \mathbb{Z}/(mn)\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ is an isomorphism.

Apparently, we can add a another equivalent statement saying that "$\mathbb{Z}/(mn)\mathbb{Z}$ and $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ are isomorphic as rings", but I have no idea about rings right now, so will wait.

We can visualize $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ as a "discrete" torus, and the homomorphism from $\mathbb{Z}/(mn)\mathbb{Z}$ as a trajectory on the torus. When m and n are coprime, the trajectory hits all points on the discrete torus.



See https://www.desmos.com/3d/qol3oxiv9e.